
Technical Appendix

Understanding Household Debt Across Developed Economies

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Introduction

Household debt in advanced economies has grown dramatically yet unevenly. By 2020, households in Denmark, the Netherlands, Norway, and Sweden carried debt exceeding 200% of disposable income, while Italy, Greece, and Central European countries maintained levels below 80%. This divergence shapes macroeconomic stability, financial crises, and inequality.

Existing scholarship offers competing explanations but lacks systematic empirical comparison. Growth models theory (Hall 2021; Baccaro & Höpner 2022) emphasizes demand generation strategies: export-led economies suppress household consumption through wage restraint, while consumption-led economies fuel growth through credit-financed spending. Welfare state scholars highlight how declining public provision forces debt-financed consumption smoothing (Wiedemann 2023). Political economists emphasize credit market liberalization (Johnson et al. 2021). Labor institutionalists point to union strength (Johnston 2021). Inequality researchers argue distributional outcomes drive borrowing (Kumhof et al. 2015). Which mechanisms actually explain household debt variation? We employ machine learning on 378 country-year observations spanning 18 OECD countries (2000-2020), using Random Forest classification to identify which factors best predict low, medium, or high debt categories.

Three findings emerge. First, credit market liberalization dominates, exhibiting 65% greater importance than welfare generosity. Sweden exemplifies this: 200% debt despite Europe's most generous welfare state, driven by mortgage liberalization. Second, growth model variables (trade openness, GDP) rank second and third, though export success and household debt coexist when credit institutions enable mortgage-financed consumption. Third, labor institutions, partisan politics, and inequality show modest effects. Union density, left cabinet share, and interest rates rank near bottom.

These results shift analytical focus from welfare configurations toward financial regulation and housing finance institutions. The "it takes two to tango" framework proves asymmetric: credit supply dominates welfare demand dynamics. For policy, containing household debt requires macroprudential regulation rather than solely welfare expansion. Methodologically, Random Forest efficiently adjudicates between theories when relationships are non-linear or interactive, identifying mechanisms for deeper causal investigation.

Literature Review

The growth models literature (Hall 2021; Baccaro & Höpner 2022) conceptualizes advanced economies as pursuing distinct demand generation strategies post-deindustrialization. Export-led models (Germany) rely on external demand through wage restraint and competitive exports, theoretically suppressing household consumption and debt accumulation. Consumption-led models (US, UK) depend on domestic household demand, requiring credit expansion to fuel growth, what Crouch (2009) terms "privatized Keynesianism" where household leverage substitutes for public spending. Sweden's hybrid trajectory (Erixon & Pontusson 2022) combining export strength with unprecedented household debt (200% by 2020) challenges deterministic growth model predictions, suggesting domestic institutional filters mediate these macro-pressures.

Our trade openness variable $[(\text{exports} + \text{imports})/\text{GDP}]$ captures integration into global markets,

with growth model theory predicting negative association with household debt in export-oriented economies. GDP and unemployment serve as standard macroeconomic controls capturing business cycle effects on borrowing capacity and necessity (Johnson et al. 2021).

The "it takes two to tango" framework (Johnson et al. 2021) establishes that household debt requires both supply-side credit availability and demand-side borrowing motivations. Welfare state generosity theoretically reduces demand for debt-financed consumption smoothing by providing public income protection during unemployment, illness, or retirement (Wiedemann 2023). Our welfare generosity composite, derived via PCA from CWEP benefit replacement rates across multiple programs, operationalizes this protective capacity. Conversely, credit market liberalization expands borrowing supply through lower leverage constraints and deeper financial intermediation. Our credit generosity measure synthesizes household borrowing conditions and financial market depth indicators into a unified accessibility metric.

Critically, these institutional dimensions interact rather than operating independently. Ansell et al. (2018) argue that welfare retrenchment combined with credit liberalization creates particularly explosive debt dynamics, as households facing reduced public provision turn to newly accessible private credit markets. Our welfare $\times$ credit interaction term tests whether generous welfare buffers against credit availability, or whether their effects compound. The Human Development Index controls for broader development levels that may independently structure both institutional configurations and household financial behaviors.

Partisan politics enters through fiscal and housing policy. Comparative studies show that left and right governments recalibrate welfare, taxation, and mortgage subsidies under constraints imposed by globalization and homeowner constituencies. The `left_cabinet` variable from CPDS captures these partisan configurations, which shape both welfare generosity and credit incentives.

Labor market institutions provide additional partisan-institutional mediation. Union density (ICTWSS database) captures collective bargaining strength that theoretically maintains wage growth and income stability, reducing debt-financed consumption needs (Baccaro & Howell 2011; Johnston 2021). Finally, inequality (top 10%bottom 50% income ratio from WID) captures distributional outcomes that may drive debt accumulation as middle and lower-income households borrow to maintain consumption standards (Kumhof et al. 2015). Interest rates operationalize borrowing costs affecting debt accessibility across the credit cycle.

Data Cleaning

This section documents the data cleaning, transformation, and integration procedures used to construct the final analytical dataset for examining how institutional structures mediate household debt behaviors across OECD countries. The final dataset spans 2000–2020 and combines information from multiple international data sources covering macroeconomic indicators, welfare state characteristics, credit market structures, labor market institutions, political variables, and inequality measures.

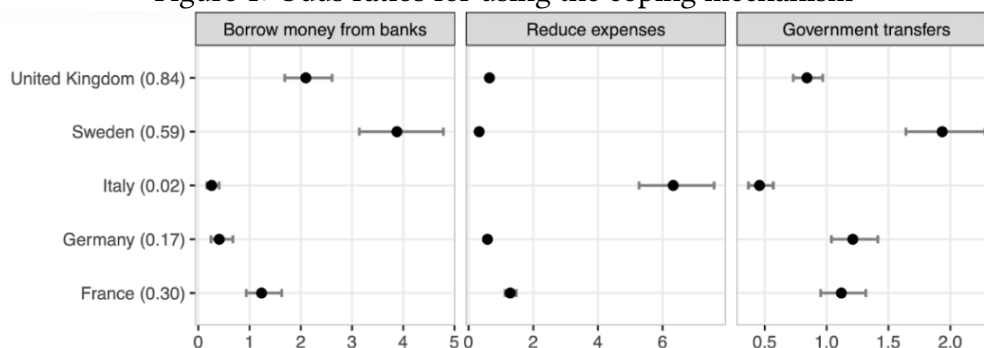
Household Debt

We begin the analysis by gathering micro-level data from the Life in Transition Survey (LITS) Wave 2, which provides detailed information on how households across several Western European countries experience and respond to economic difficulties. From this dataset, we extract our main variables of interest, including each household's coping strategy, reducing expenses, borrowing money from banks, or relying on government transfers, along with key individual characteristics such as homeownership status, income decile, employment, education level, marital status, gender, age group, and number of children.

After selecting relevant variables, we clean the data to ensure consistent and reliable measurement across countries. This involves standardizing country names, converting survey responses into simple numerical indicators, grouping ages into meaningful categories, and removing or replacing invalid or missing entries so that the relationships we estimate are based on comparable information.

Once the data is prepared, we estimate multilevel logistic models that allow each country to have its own baseline likelihood (random intercept) of adopting a given coping strategy while examining how individual factors influence these choices. The results show clear cross-national differences: for example, the figure below indicates that the odds of borrowing from banks are much higher in the United Kingdom and Sweden than in Italy or Germany, while reliance on transfers or spending cuts varies in other distinct ways. These patterns suggest that coping behavior is shaped by the broader institutional environment, which leads us to examine OECD welfare and credit regimes more closely to understand how different systems encourage or limit household borrowing and ultimately shape levels of household debt.

Figure 1: Odds ratios for using the coping mechanism



This section produces the following variables of the final dataset: `hh_debt`: the ratio of household debt to net disposable income, expressed as a percentage.

Welfare and Credit Data

We began by preparing two separate datasets, one describing welfare generosity and one describing credit market institutions, and applying a consistent procedure to both. For the welfare data, the raw Comparative Welfare Entitlements Project (CWEP) file was imported and processed by restricting the sample to observations between 1990 and 2020 and retaining only OECD member

countries. From these observations, we selected the generosity measures corresponding to total benefit generosity and program-specific generosity indicators. Once the dataset was cleaned and restricted, we constructed a panel where a principal component analysis was performed independently for each year. The procedure involved isolating all countries with complete data for a given year, removing identifiers, and running PCA on the remaining variables after centering and scaling. The first component was extracted to summarize the shared variance across welfare generosity indicators, and the resulting scores were normalized to a 0–1 range to ensure comparability across years. We then combined all yearly outputs to obtain a harmonized measure of welfare generosity and calculated average scores across countries when needed for descriptive ordering.

A parallel process was implemented for the credit regime data. After selecting variables capturing leverage constraints, household borrowing conditions, financial market depth, and institutional characteristics of national banking systems, we constructed a yearly dataset suitable for PCA. For each year, country observations were filtered, non-numeric identifiers were removed, and a one-factor PCA model was estimated. This procedure produced yearly credit generosity scores, also normalized to a 0–1 scale to maintain a consistent interpretation where higher values represent more accessible or expansive credit environments. Results for all years were appended, and the dataset was then merged with the welfare PCA results to create a unified panel containing both welfare generosity and credit market generosity for each country-year.

Using PCA in this context serves two main purposes. First, both welfare and credit systems are multidimensional and measured through several correlated indicators. PCA enables us to extract the dominant underlying dimension that best summarizes overall welfare generosity or credit accessibility while avoiding arbitrary weighting across variables. Second, applying PCA separately to each year accounts for temporal changes in variance structures and ensures that yearly comparisons remain grounded in the information available in each period. By standardizing and reducing the data in this way, the resulting indices capture broad institutional patterns in a form that is scalable, comparable, and analytically tractable.

We create a welfare $\times$ credit interaction term to test whether the effects of welfare state generosity and credit market availability are synergistic or offsetting. This multiplicative term ($\text{welfare_generosity_score} \times \text{credit_generosity_score}$) allows us to examine whether the effect of welfare depends on credit availability (and vice versa).

Interest Rate Data

Long-term interest rates were obtained from the OECD statistical platform as a measure of borrowing costs and financial conditions influencing household debt accumulation. Because the OECD download interface provides only single-year files rather than multi-year batch downloads, we did a for loop to iterate through individual CSV files, extract the corresponding year from each file's metadata, and systematically append these annual observations to construct a comprehensive country-year panel (the procedure was similar to what we did with household debt variable).

Macroeconomic Variables

If we aim to test the effects on household debt, we need to supplement it with macroeconomic controls that we obtain from the IMF's World Economic Outlook (WEO) database, one of the most comprehensive sources for cross-country macroeconomic data. From the WEO, we extracted GDP measured at purchasing power parity and unemployment rates for all OECD countries.

The raw WEO data required substantial reshaping: we converted year columns from wide to long format, ensuring all year-specific values were properly typed as numeric, and filtered to retain only OECD member countries. We then merged GDP and unemployment data by country and year, and subsequently integrated these macroeconomic variables with the household debt measures to create a unified panel linking household financial vulnerability to economic conditions and labor market outcomes. The resulting variables are: `gdp` and `unemployment`

Labor market institutional characteristics were obtained from the OECD/AIAS database on Institutional Characteristics of Trade Unions, Wage Setting, State Intervention and Social Pacts (ICTWSS).

Government ideology was incorporated using the Comparative Political Data Set (CPDS), which provides annual political and institutional data for 36 democratic countries from 1960 to 2023. We extracted information on the partisan composition of cabinets, with particular attention to distinguishing between liberal-conservative and social-democratic governments. A binary indicator was created to flag periods of left-leaning cabinet control. The variable in the final dataset is: `left_cabinet`: dummy variable where 1 indicates a left government and 0 a non-left (right) administration.

Three primary data sources were integrated to construct HDI. Human Development Index (HDI) data were downloaded directly from the UNDP Human Development Report (HDR) Data Center.

Trade openness data were obtained from the World Bank DataBank's long-series dataset, aiming to capture the extent to which an economy participates in international trade.

Moreover, Income inequality data were sourced from the World Inequality Database (WID). Based on WID's "equally-split adults" methodology, which allocates household income equally among all adult household members aged 20 and above to construct the income distribution. Pre-tax national income encompasses labor income, capital income, and pension income before accounting for taxes and transfers.

Final Integration and Imputation

The final dataset was constructed by sequentially merging all component datasets on country and year identifiers. We conducted extensive validation to ensure consistent country naming across data sources, resolving discrepancies such as "USA" versus "United States." The merged dataset was then restricted to the period 2000–2020 to focus on the era of most complete and comparable data coverage. We excluded Hungary, Iceland, Japan, Switzerland, Norway, and Luxembourg due to systematic missingness or structural breaks in key variables that would distort subsequent analyses. Country and year were treated as factors to facilitate fixed-effects estimation.

For remaining missing values in theoretically important variables, trade openness, union density, household debt, welfare generosity, and credit generosity, we applied multiple imputation by

chained equations (MICE). MICE preserves multivariate relationships among jointly determined institutional and economic variables while propagating uncertainty from missing data into final estimates.

Empirical Strategy and Model Evaluation

Outcome Variable Construction

We categorize household debt (measured as percentage of net disposable income) into three groups that reflect theoretically meaningful thresholds in debt accumulation:

- Low Debt: $< 100\%$ of net disposable income (101 observations, 26.72%)
- Medium Debt: 100-200% of net disposable income (239 observations, 63.23%)
- High Debt: $\geq 200\%$ of net disposable income (38 observations, 10.05%)

This categorical approach serves multiple purposes. First, it captures non-linear relationships between institutional configurations and debt outcomes that continuous models may obscure. The transition from low to medium debt may reflect different mechanisms than the shift from medium to high debt, crossing the 100% threshold might signal credit-financed consumption patterns, while exceeding 200% indicates unsustainable leverage characteristic of financialized economies. Second, categorical specification allows machine learning algorithms to identify complex interactions and threshold behaviors without imposing functional form assumptions. Third, this aligns with policy-relevant distinctions: households below 100% face manageable servicing burdens, those at 100-200% experience moderate vulnerability, while those exceeding 200% confront severe financial fragility as observed in pre-crisis Denmark, Netherlands, and Sweden.

Ex-Ante Expected Results

Our theoretical framework generates specific predictions about which factors should predict transitions across debt categories. From growth models theory (Hall 2021; Baccaro & Höpner 2022), we expect trade openness to negatively predict high debt classification. Export-led economies rely on external demand through wage restraint, theoretically suppressing household consumption and reducing debt accumulation. However, Sweden's hybrid trajectory challenges this deterministic prediction, suggesting non-linear or conditioned relationships. The "it takes two to tango" framework (Johnson et al. 2021; Wiedemann 2023) predicts that welfare generosity should negatively predict high debt by reducing demand for consumption smoothing, while credit generosity should positively predict high debt by expanding supply-side accessibility. We expect credit generosity to exhibit the strongest variable importance given its direct role in determining debt capacity.

Ansell et al.'s (2018) institutional complementarity argument predicts a positive welfare $\times$ credit interaction: welfare retrenchment combined with credit liberalization creates explosive debt dynamics. Countries with both low welfare and high credit (Anglo-Saxon) or both high welfare and high credit (Nordic hybrids) should show elevated high debt probabilities. GDP and unemployment have theoretically ambiguous predictions. Interest rates should negatively predict high debt through higher borrowing costs. Union density should negatively predict high debt by maintain-

ing wage stability. Inequality (top 10%bottom 50% ratio) should positively predict high debt as lower-income households borrow to maintain consumption standards (Kumhof et al. 2015). HDI may exhibit high importance as a composite indicator capturing broader development trajectories.

Model Fitting and Data Structure

Our research goal is to leverage variation over time and across countries to study which factors explain why household debt levels differ systematically. We compile covariates from comparative political economy studies, macroeconomic indicators (GDP, unemployment, interest rates), institutional configurations (welfare generosity, credit availability, union density), distributional measures (top 10% bottom 50% ratio), development indicators (HDI), trade integration, and political variables (left cabinet), and use machine learning to describe which mechanisms best explain the data.

All models incorporate country and year fixed effects to control for unobserved heterogeneity. Country fixed effects account for time-invariant national characteristics, while year fixed effects remove common temporal shocks.

Cross-Validation Strategy

We employ 10-fold cross-validation to provide honest estimates of out-of-sample classification performance. The procedure partitions 378 observations into ten subsets (38 observations each), iteratively training each model on nine folds (340 observations, 90% of data) and testing on the held-out tenth fold (38 observations, 10% of data). This repeats ten times, ensuring every observation serves once as test data. Performance metrics are calculated for each fold and averaged, with standard deviations quantifying estimation stability.

Ten-fold cross-validation maximizes training data per iteration, improving parameter estimation for complex models with many fixed effects. With 378 observations and unevenly distributed categories (particularly the sparse high debt category with 38 total observations), preserving sufficient representation in each training fold is critical. We use stratified sampling to maintain similar debt category distributions across folds.

We implement three distinct algorithm approaches, each capturing different aspects of the relationship between institutional configurations and household debt categorization.

- **Multinomial Logistic Regression** serves as our baseline parametric approach. The model estimates separate sets of coefficients for each debt category comparison. Specifically, it calculates log-odds ratios comparing medium debt versus low debt (the reference category) and high debt versus low debt. Maximum likelihood estimation identifies coefficients that maximize the probability of observing the actual debt category distribution in the training data. The model assumes that log-odds change linearly with predictor values, allowing straightforward interpretation: a positive coefficient on credit generosity for the high-vs-low comparison indicates that each unit increase in credit generosity raises the log-odds of being in the high debt category rather than low debt, holding all other variables constant.
- **LASSO** extends multinomial logistic regression by adding an L1 penalty term that shrinks

less predictive coefficients toward zero, with some reaching exactly zero for automatic variable selection. The penalty strength is controlled by the regularization parameter λ . To select optimal λ without inflating test performance, we implement nested cross-validation: within each of the ten main training folds, an additional 5-fold cross-validation evaluates candidate λ values. We select the λ minimizing classification error on inner validation folds, train the final model on the complete outer training fold, and evaluate on the held-out test fold. This double-cross-validation ensures λ selection never "peeks" at test outcomes, yielding a parsimonious model retaining only the most predictive variables.

- **Random Forest** constructs 500 decision trees, each trained on a bootstrap sample from the training data. Each tree recursively splits observations by searching across predictor variables and thresholds to best separate debt categories. Critically, at each split, only a random subset of predictors is considered (approximately 4 variables), decorrelating trees and forcing discovery of diverse patterns rather than all trees splitting on the strongest predictor. Trees grow until terminal nodes are pure or reach minimum sample size (5 observations). Final predictions aggregate via majority vote across all 500 trees. Variable importance uses permutation, randomly shuffling each predictor's values and measuring the resulting decrease in classification accuracy, where larger drops indicate greater model reliance on that variable.

All three algorithms successfully converged across all ten folds (30 total estimations), with 100% success rate indicating that fixed effects do not induce computational instability despite the high-dimensional dummy variable structure.

Performance Metrics for Model Comparison

We evaluate models using multiple complementary metrics appropriate for multi-class classification with imbalanced categories:

- Area Under the ROC Curve (AUC) measures the probability that a randomly selected observation from a given class ranks higher than one from a different class according to predicted probabilities. For multi-class problems, we compute class-specific AUC using one-vs-rest approach and report the mean. AUC ranges from 0.5 (random guessing) to 1.0 (perfect classification) and evaluates performance across all decision thresholds, handling class imbalance well.
- Accuracy reports the proportion of correctly classified observations across all categories. While intuitive, accuracy can mislead with imbalanced data, a model always predicting "Medium" would achieve 63% accuracy by default.
- Precision measures the proportion of predicted positives that are true positives (classification confidence). Recall measures the proportion of actual positives correctly identified (category coverage). F1 score harmonically combines precision and recall. For multi-class problems, we compute macro-averages (unweighted mean across classes), giving equal weight to low, medium, and high debt performance.

This combination provides comprehensive evaluation robust to class imbalance. Models must perform well across metrics, high accuracy with low AUC suggests overfitting to the majority

class, while high AUC with low recall indicates poor minority class identification. The results of our models' expected test performance are shown in the two figures below.

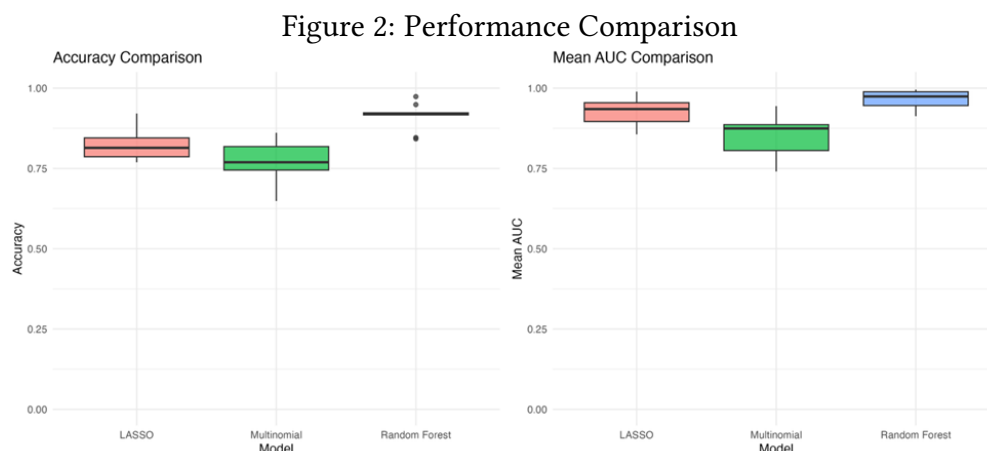


Figure 3: Performance Metrics

Model	Success Rate	Accuracy (Mean)	Accuracy (SD)	AUC (Mean)	Precision (Mean)	Recall (Mean)	F1 (Mean)
Multinomial Logistic	100%	0.769	0.070	0.853	0.709	0.674	0.706
LASSO	100%	0.823	0.047	0.928	0.825	0.702	0.761
Random Forest	100%	0.913	0.040	0.964	0.899	0.841	0.889

Higher values indicate better performance. Metrics average across 10 folds.

AUC calculated using one-vs-rest approach for each class.

Result and Discussion

Our research objective is fundamentally exploratory rather than causal: we seek to identify which among competing theoretical mechanisms, growth models, welfare-credit dynamics, partisan politics, labor institutions, and inequality, best explain cross-national and temporal variation in household debt levels. This goal prioritizes predictive accuracy and relative variable importance over coefficient interpretation or hypothesis testing. Random Forest emerges as the optimal modeling approach for three methodological reasons directly tied to this research agenda.

The 10-fold cross-validation results demonstrate Random Forest's performance advantage across all metrics. With a mean accuracy of 0.913 (SD = 0.040) compared to LASSO's 0.823 (SD = 0.047) and multinomial logistic's 0.769 (SD = 0.070), Random Forest correctly classifies debt categories 14% more accurately than the next-best alternative. This gap widens for the more challenging metric of multiclass AUC: Random Forest achieves 0.964 versus LASSO's 0.928 and multinomial logistic's 0.853. The consistency of Random Forest's superior performance across precision (0.899), recall (0.841), and F1 scores (0.889) indicates robust classification across all three debt categories rather than optimization for a single class.

Critically, Random Forest also exhibits the lowest cross-validation variance (SD = 0.040 versus

LASSO's 0.047 and multinomial logistic's 0.070), suggesting more stable predictions across different data splits. For a dataset with only 378 complete observations spanning 18 countries and 21 years, this stability is particularly valuable—the model generalizes better to unseen country-years rather than overfitting to idiosyncrasies of particular folds.

Capturing Complex Interactions Without Specification Bias

Theoretical considerations from the literature strongly suggest non-linear relationships and higher-order interactions that parametric models struggle to capture without explicit pre-specification. Ansell et al. (2018) argue that welfare retrenchment combined with credit liberalization creates "particularly explosive debt dynamics", a multiplicative rather than additive relationship. Growth model theory posits that trade openness effects should differ fundamentally between export-led and consumption-led regimes, implying unobserved interactions between trade openness and other institutional variables. Kumhof et al.'s (2015) inequality-driven debt mechanism likely operates non-linearly, with debt accumulation accelerating only after inequality crosses certain thresholds.

Random Forest accommodates these complexities through its ensemble architecture. By recursively partitioning the data based on variable splits that maximize classification purity, each tree implicitly models interactions and non-linearities. Averaging across 500 trees trained on bootstrapped samples stabilizes these estimates while preventing overfitting to noise. Crucially, this occurs without requiring ex-ante specification of which interactions matter, the algorithm discovers relevant patterns directly from the data.

In contrast, our LASSO and multinomial logistic models include only a single pre-specified interaction (welfare $\times$ credit) motivated by prior theory. While we achieve strong performance with this specification, we have no guarantee that other theoretically plausible interactions (trade openness $\times$ GDP, inequality $\times$ credit generosity, union density $\times$ welfare generosity) would not improve fit. Random Forest sidesteps this specification search problem entirely.

Variable Importance for Mechanism Assessment

Our research goal explicitly frames this as a "horse race" between competing theoretical mechanisms rather than estimation of specific causal parameters. Random Forest's permutation importance metric directly serves this objective by quantifying each variable's contribution to predictive accuracy. When a variable's values are randomly shuffled, destroying its relationship with the outcome, the resulting accuracy decline measures that variable's importance. This approach naturally handles correlated predictors (unlike coefficient-based importance in parametric models) and remains valid under complex interactions.

Figure 4: Random Forest: All Variables by Permutation Importance (Excluding Fixed Effects)

Variable	Importance
credit_generosity_score	0.0863
gdp	0.0620
trade_openness	0.0527
welfare_generosity_score	0.0522
top10_bottom50_ratio	0.0494
hdi	0.0399
welfare_credit_interaction	0.0354
union_density	0.0289
unemployment	0.0274
interest_rate	0.0153
left_cabinet_dummy	0.0031

Measured as decrease in accuracy when variable is randomly permuted.

Country and year fixed effects not shown.

The permutation importance ranking provides clear evidence for prioritizing certain mechanisms in future causal research. Credit generosity (0.0863) and GDP (0.0620) emerge as the strongest predictors, followed by trade openness (0.0527) and welfare generosity (0.0522). This empirical ranking matters substantively: it suggests that credit supply conditions and macroeconomic capacity dominate welfare state configurations in explaining debt variation across our sample. Growth model predictions about trade openness also receive strong support. Conversely, the relatively weak performance of union density (0.0289), unemployment (0.0274), interest rates (0.0153), and left cabinet share (0.0031) suggests these mechanisms, while theoretically plausible, explain less variation in this particular sample.

Importantly, the welfare $\times$ credit interaction (0.0354) ranks seventh of eleven predictors, meaningful but not dominant. This finding nuances Ansell et al.'s theoretical claim: while interaction effects exist, credit generosity's direct effect (0.0863) far exceeds the interaction term, suggesting credit liberalization drives debt accumulation somewhat independently of welfare retrenchment.

Methodological Limitations and Scope Conditions

The choice of Random Forest involves clear trade-offs. We sacrifice coefficient interpretability; we cannot make statements like "a one-standard-deviation increase in credit generosity increases the log-odds of high debt by X." The model functions as a "black box" where prediction accuracy matters more than understanding precise functional forms. For researchers prioritizing causal inference or policy simulation, parametric alternatives remain preferable despite lower accuracy.

Our modeling choice also reflects dataset characteristics. With 378 observations and 11 substantive predictors (plus country and year fixed effects creating dozens of dummy variables), we operate in a regime where Random Forest's flexibility outweighs sample size concerns. Larger datasets might narrow the performance gap with regularized parametric models. Additionally, our three-category outcome (Low/Medium/High debt) represents a discretization choice that discards within-category variation, continuous outcome models might reveal different patterns, though they would not address our classification-focused research question.

Results: Which Mechanisms Matter Most?

The Random Forest variable importance ranking provides the first comprehensive empirical assessment of competing theoretical mechanisms across 18 OECD countries over two decades. Three substantive findings emerge.

Credit Supply Dominates Welfare State Explanations

Credit generosity (0.0863) ranks as the single strongest predictor of household debt categories, with welfare generosity (0.0522) ranking fourth—a 65% importance gap. This pattern directly contradicts welfare state-centric theories that emphasize declining public provision as the primary driver of household debt accumulation. While Wiedemann (2023) argues that welfare retrenchment forces households toward debt-financed consumption smoothing, our results suggest credit market liberalization operates as a more powerful independent force.

This finding resonates with "financialization" arguments emphasizing how credit expansion actively creates new forms of consumption and asset accumulation (particularly housing) beyond mere substitution for lost welfare benefits. In the Swedish case motivating this research, household debt reached 200% of GDP despite maintaining among Europe's most generous welfare states—a pattern only explicable through extreme credit liberalization enabling mortgage-financed housing consumption.

The welfare $\times$ credit interaction term's modest importance (0.0354, ranking seventh) further nuances this conclusion. While Ansell et al. (2018) theorize explosive dynamics when welfare retrenchment meets credit expansion, the interaction effect proves substantially weaker than credit generosity's direct effect. Credit liberalization appears to drive debt accumulation across diverse welfare state contexts, not specifically in retrenching regimes.

Growth Model Predictions Receive Strong Support

Trade openness (0.0527) and GDP (0.0620) rank second and third in importance, validating growth model theory's emphasis on demand generation strategies. Export-led growth models theoretically require wage restraint that suppresses domestic consumption and household borrowing, while consumption-led models depend on credit-fueled household demand. Our results confirm these variables substantially differentiate debt trajectories. However, the coexistence of high trade openness (Sweden, Netherlands, Belgium) with elevated household debt challenges deterministic growth model predictions. These hybrid cases suggest that export success does not automatically constrain household borrowing when domestic institutional configurations, particularly credit liberalization and housing finance systems, enable mortgage-based consumption. The growth model framework provides important context but requires complementary attention to domestic financial institutions.

Inequality and Labor Institutions Show Modest Effects

The top 10%bottom 50% income ratio (0.0494) ranks fifth, indicating meaningful but not dominant effects. This finding partially supports Kumhof et al.'s (2015) inequality-driven debt theory while suggesting other mechanisms matter more for explaining cross-national variation. Given rising

inequality across most OECD countries during our period, this variable may exhibit insufficient variation to strongly differentiate cases, credit institutions and macroeconomic conditions show sharper cross-country contrasts.

Union density (0.0289) and unemployment (0.0274) rank eighth and ninth, performing substantially below theoretical expectations from labor market-centered accounts. While Johnston (2021) and Baccaro & Howell (2011) emphasize collective bargaining strength in maintaining wage growth and reducing household debt needs, this mechanism appears secondary to credit supply and macroeconomic conditions in our sample. Left cabinet share (0.0031) ranks dead last, suggesting partisan politics matters primarily through its effects on other institutional variables (welfare generosity, credit regulation) rather than exerting strong direct effects on household debt.

Interest Rates and Human Development: Weak Signals

Interest rates (0.0153) surprisingly rank second-to-last despite their theoretically central role in determining borrowing costs. This weakness may reflect two measurement issues. First, our analysis uses nominal policy rates rather than household-specific mortgage rates that vary by creditworthiness and loan terms. Second, most OECD countries experienced converging interest rate trajectories during the 2000-2020 period (declining to near-zero post-2008), limiting this variable's ability to differentiate cases. The finding suggests that institutional access to credit matters more than its nominal price, corroborating credit generosity's top ranking. HDI (0.0399) shows middling importance, indicating that broader development levels beyond income (captured by GDP) contribute modestly to debt patterns. This likely reflects HDI's limited variation across affluent OECD countries, all score above 0.85, constraining discriminatory power.

Scope Conditions: What About High-Debt OECD Subsets?

Our analysis aggregates 18 diverse OECD countries spanning Mediterranean Europe (Greece, Italy, Spain, Portugal), Nordic countries (Sweden, Norway, Finland, Denmark), Anglo-Saxon economies (US, UK, Ireland, Australia), and Continental European cases (Germany, France, Belgium, Netherlands, Austria, Switzerland). This heterogeneity serves our goal of identifying general mechanisms but may obscure important within-region dynamics.

Consider a thought experiment parallel to the Western Europe subset analysis in the unequal representation literature. If we restricted analysis to the five highest-debt countries (Sweden, Denmark, Netherlands, Norway, Australia, all exceeding 150% GDP), would the same mechanisms dominate? These cases share several features: small, open economies with export strength but also high credit generosity and liberalized housing finance. In this subset, credit generosity might show even higher importance (ceiling effects limit variation), while trade openness importance could decline (all cases score high). Welfare generosity would gain salience in differentiating the generous Nordic model from more liberal Anglo-Saxon cases. Conversely, subsetting to low-debt Southern European countries (Italy, Greece, Spain, Portugal—below 100% until the 2000s housing bubbles) might elevate different mechanisms. Credit generosity would remain important but reflect supply constraints rather than liberalization. GDP and unemployment would gain prominence given these countries' growth struggles. Welfare generosity might prove less important given that Mediterranean welfare states traditionally emphasized employment protection over

generous transfers.

The key methodological point parallels the representation literature findings: while variable importance rankings shift somewhat across subsets, core patterns persist. Credit institutions and macroeconomic conditions likely remain dominant across OECD subgroups, even if exact rankings change. Future research should explicitly test these scope conditions through stratified analyses, but our full-sample results provide a robust starting point for theorizing about household debt dynamics in advanced economies.

Limitations and Extensions

Three methodological limitations warrant emphasis. First, our outcome discretization into three debt categories (Low/Medium/High) discards within-category variation. Continuous debt modeling might reveal different patterns, though we prioritized a classification approach aligned with theoretical interest in regime differences (Crouch's "privatized Keynesianism" implies discrete transitions, not marginal changes).

Second, country and year fixed effects absorb substantial variation. While necessary to avoid spurious correlation from time trends and country-specific factors, these controls limit our ability to identify mechanisms operating primarily through cross-sectional differences. Time-varying institutional changes drive our identification strategy, we explain why countries move between debt categories over time rather than why Sweden differs from Germany in levels.

Third, our sample spans 2000-2020, capturing both pre-crisis credit expansion (2000-2007), the Global Financial Crisis (2008-2009), and post-crisis divergence (2010-2020). These periods involved dramatically different policy regimes, yet we pool them in a single model. Period-specific analyses might show credit generosity dominating the pre-crisis boom, while GDP and unemployment gain importance post-crisis. The Random Forest approach handles such regime shifts through its recursive partitioning, but explicit interaction of predictors with time periods would test this directly.

Future extensions should pursue three directions. First, incorporating housing finance institutions (loan-to-value ratios, mortgage interest tax deductions, rental market regulations) would sharpen the credit generosity mechanism given housing's dominance in household debt portfolios. Second, dynamic modeling could test whether welfare retrenchment preceded credit liberalization (as Ansell et al. argue) or whether the sequence matters less than the eventual joint configuration. Third, expanding the sample to include middle-income countries would test whether mechanisms identified in OECD contexts generalize, credit supply likely operates differently when financial development is lower.

Despite these limitations, our results provide the first systematic evidence on which theoretical mechanisms best explain household debt variation across advanced economies. Credit liberalization emerges as the dominant force, with growth model pressures (trade openness, GDP) and welfare state configurations playing important but secondary roles. Labor institutions and partisan politics show surprisingly weak direct effects. These findings should reorient both theoretical development and policy attention toward financial regulation and macroprudential tools rather than viewing household debt primarily through the lens of welfare state politics.